

# Time Series

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## Todo.

- look at M3 forecast challenge
- judge forecast distributions
- linear regression model (Matrix notation). Also show std and cross validation
- state space models and innovations state space models.
- model selection (BIC, AIC, AICc)
- Exponential smoothing

**Introduction.** Time series are commonly thought as a superposition of several components

- *Level*: The central value or baseline around which the time series data varies.
- *Trend*: long-term direction of time series
- *Seasonal*: pattern that repeats with known and fixed periodicity
- *Cycle*: pattern that repeats but with unknown und changing periodicity.
- *Autocorrelation*:
- *Error*: unpredictable component of time series.

Many time-series models can be classified as additive, multiplicative, or a combination of these components [3]. The aim to time series modelling is to capture all effects until only error (=noise) is left. More precisely if there is still some uncaptured trend, seasonal, cycle or autocorrelation then we can be pretty sure that we can improve the model. However, this does not imply that all models that just exhibit unexplained noise perform equally.

**Autocorrelation function (ACF)** The autocorrelation function (ACF) is the Pearson correlation of a time dependent random variable  $Y_t$  with itself shifted in time by  $\tau$ . The ACF is typically computed on the raw data (to observed lag relationships) and residuals (for model diagnostic).

$$\rho(\tau) = \frac{\sum_{t=\tau+1}^T (y_t - \bar{y})(y_{t-\tau} - \bar{y})}{\sum_{t=1}^T (y_t - \bar{y})^2}, \quad (1)$$

where  $\bar{y}$  is the (empirical) mean. The autocorrelation function depends on  $\tau$ . For white noise 95 % of its ( $\tau$ -dependent) values are within the interval  $\pm 2/\sqrt{T}$ . Thus the ACF is used to identify white noise [6].

Datasets with trend tend to have large, positive autocorrelation at small lags. Seasonal data have larger autocorrelations at seasonal lags (and multiples of it). Upon model diagnostics ACF is often computed for the residuals.

**Stationary time series** A stationary time series has a pdf that is invariant under time translation. It is a requirement for ARIMA models.

**Definition 1** (Stationary time series). *let  $\{X_t\}$  be a stochastic process and let  $F_X(x_{t_1+\tau}, \dots, x_{t_n+\tau})$  represent the cumulative distribution function of the unconditional joint distribution. Then  $\{X_t\}$  is said to be **strictly stationary**, if*

$$\begin{aligned} F_X(x_{t_1+\tau}, \dots, x_{t_n+\tau}) &= F_X(x_{t_1}, \dots, x_{t_n}) \\ \forall \tau, t_1, \dots, t_n &\in \mathbb{R}, \\ \forall n &\in \mathbb{N}_{>0} \end{aligned} \quad (2)$$

*It is said to be **N-th order stationary** if*

$$\begin{aligned} F_X(x_{t_1+\tau}, \dots, x_{t_n+\tau}) &= F_X(x_{t_1}, \dots, x_{t_n}) \\ \forall \tau, t_1, \dots, t_n &\in \mathbb{R}, \\ \forall n &\in \{1, \dots, N\} \end{aligned} \quad (3)$$

So N-th order stationarity is a weaker condition. A non-stationary time series can be made stationary by 1) differencing, 2.) log transformation (make variance uniform), 3.) seasonal differencing, 4.) de-trending,

**Cross validation.** For time series a split in test and train data is performed along the temporal axis. Observations in the past –on which the model is trained – are called train set and observations in the futures –for which the accuracy is determined – are called test set.

A more sophisticated split of data is called cross validation. In this approach there is a series of train/test sets. Each of which contains only one observation  $y_l$  in the test set and the training set are the observations *prior* to this observation with an offset ("step")  $s$   $y_1, \dots, y_{l-s}$ . For fixed  $s$ ,  $l$  induces a series of test/train sets. The forecast accuracy is then measured by averaging over the test sets. This is also called *evaluation on a rolling forecast origin*.  $s = 1$  is called 1-step-forecast. In practice the accuracy as a function of  $s$  is investigated.

## 1 Model diagnostics.

Assume that there is model that makes predictions  $\hat{y}_t$  at time  $t$  and the corresponding observed values  $y_t$  are known. Then there are several methods that help diagnosing the model: residuals, autocorrelation function, and portmonaie tests.

**Residuals.** Residuals of a time series is the difference between predicted values  $\hat{y}_t$  and observed values  $y_t$  at time  $t$ ,

$$e_t = y_t - \hat{y}_t. \quad (4)$$

If a transformation was applied to the data (e.g., the Box-Cox transformation) then the residuals computed on the transformed space are called *innovative residuals*. Innovative residuals should be used to judge a model.

The residuals of "good" model should be *white noise*, e.g. there is no further information left in the residuals. This means:

- Zero mean
- Uncorrelated
- Normally distributed with constant variance (homoscedasticity)

The latter point is beneficial to ease the computation of prediction intervals (for certain models). If the mean is not zero then the model is biased and if there is correlation then there is still some information left in the residuals that should go in the model [6].

These criteria do not assist in choosing a model as they may hold for several models simultaneously (model selection is performed via the error on the test set). Fixing the correlation and the distributional shape of the residual is challenging whereas zero mean can be achieved by adding a constant (the mean) to the model. A Box-Cox transformation may achieve a normal constant variance distribution.

**Portemanteau tests for ACF** perform hypothesis tests for the ACF as a point wise analysis of  $\rho(\tau)$  for each  $\tau$  could yield false positives (claiming erroneous autocorrelation). These tests are based on statistics of the ACF that are  $\chi^2$  distributed for white noise (this enables a hypothesis test with the null hypotheses that a ACF is white noise).

- The **Box-Pierce** test is based on the statistics (for fixed  $\ell$ )

$$Q = T \sum_{\tau=1}^{\ell} \rho_{\tau}^2 \quad (5)$$

- The more accurate **Ljung-Box** test is based on the statistics (for fixed  $\ell$ )

$$Q = T(T+2) \sum_{\tau=1}^{\ell} \frac{\rho_{\tau}^2}{T-\tau} \quad (6)$$

For both tests, large values of  $Q$  indicate that the ACF does *not* come from white noise. More precisely, if the autocorrelations are from white noise then  $Q$  would have a  $\chi^2$  distribution with  $\ell - K$  degrees of freedom, where  $K$  is the number of parameters in the model (so, if  $Q$  is calculated on the raw data rather than the residuals  $K = 0$ ).

In practice, the **choice**  $\ell = 10$  for non-seasonal data and  $\ell = 2p$  where  $p$  is the period of seasonality is common. For large  $\ell$  (compared to  $T$ ) the test deteriorates, which suggests the rule of thumb  $\ell = \min(10, T/5)$  ( $\ell = \min(2p, T/5)$ ) for non-seasonal (seasonal) data [6].

**Roundtrip.** A typical model diagnosis consists of

- Plot innovative residuals: (zero mean and constant variance)
- Plot ACF together with Ljung-Box Test
- Plot histogram of residuals (normally distributed). Look at shape and outlier.

From a practitioner's point of view, if a model's residual turns out not to be white noise with normal, constant variance distribution, it will rather effect the prediction distribution rather than the point estimates of a refined model.

## 2 Loss functions and forecast errors

Forecast errors judge the performance on a model on unseen data. Thus they have to applied to *test* (out of sample data) data. In contrast model diagnostics based on residuals is performed on training (in sample) data. Furthermore it is important to distinguish between evaluating the point estimate and the estimated density, which gives raise to different methods

### Evaluating point forecasts

**Definition 2** (Loss functions). *Let  $y_t$  be the observed time series and  $\hat{y}_t$  the corresponding predictions at timestep  $t$  of  $T$  timesteps.*

- *Absolute loss*
- *Squared loss*
- *Huber loss*

$$\begin{cases} \frac{1}{2}(y_t - \hat{y}_t)^2 & \text{if } |y_t - \hat{y}_t| \leq \delta, \\ \delta |y_t - \hat{y}_t| - \frac{1}{2}\delta^2 & \text{otherwise.} \end{cases} \quad (7)$$

- *$\rho$ -risk metric (quantile loss) [7]*

**Definition 3** (Forecast errors). *Let  $y_t$  be the observed time series and  $\hat{y}_t$  the corresponding predictions at timestep  $t$  of  $T$  timesteps.*

- *Mean absolute error (MAE):  $\frac{1}{T} \sum_{t=1}^T |y_t - \hat{y}_t|$*
- *Mean squared error:  $\frac{1}{T} \sum_{t=1}^T (y_t - \hat{y}_t)^2$ .*
- *Root mean squared error (RMSE):  $\sqrt{\frac{1}{T} \sum_{t=1}^T (y_t - \hat{y}_t)^2}$*

A forecast method that minimizes the MAE (RMSE) will lead to forecasts of the median (expectation value) of the underlying forecast density. Thus, the RMSE is widely used although the MAE is easier to interpret. Both, the MAE and RMSE dependent on the physical units.

In order to compare algorithms among various problems unit independent measures are needed. However this is a delicate task and quantities such as the the (symmetrized) mean absolute percentage error may diverge. Rather, errors scaled by a simple forecast method are more appropriate [6, 4].

## Evaluating forecast distributions

### 3 Data transformations and patterns

**Transformations.** The aim of a transformation is to make variations of a time series homogeneous., which makes modelling and time series decomposition much easier. Simpler models can be used for transformed data, which exhibit homegenous variation.

- **Calender adjustments.** Sometimes pattern occur that are due to different number of days in a month or if there is some additional known contraint in the data (data points are not generated on sundays). It is much easier to remove this structure by changing to an appropriate time unit than trying to let the model learn this.
- **Correction for known temporal adjustments.** If data is affected by e.g. the population or inflation adjust it in order to account for these effects on the training data level instead of trying to learn these effects. E.g. consider quantities per capita (population) or use inflation corrected data (financial times series).
- **Fourier transformation.** Include the dominant Fourier contributions in order to help the model dealing with periodicity. In practice these components my be found by an FFT and sometimes only the corresponding basis functions are included.
- **Box-Cox transformation** on target values [6],

$$y_t \mapsto \begin{cases} \ln y_t & \text{if } \lambda = 0 \\ (y_t^\lambda - 1)/\lambda & \text{otherwise.} \end{cases} \quad (8)$$

This transformation aims to render the seasonal effects to be about the same across the whole time series, which dictates the choice of  $\lambda$ . So one can screen values  $\lambda = 1 \rightarrow \lambda = -1$  corresponding to weak/no to strong transformation and picking that value of  $\lambda$  for which the variation becomes homogeneous. In practice forecasting results (mean

point estimates) are relatively insensitive to  $\lambda$  in contrast prediction intervals. Once a prediction has been made on the transformed space it needs to be back-transformed into the original scale. The inverse transformation

$$y_t \mapsto \begin{cases} e^{y_t} & \text{if } \lambda = 0 \\ (\lambda y_t + 1)^{\frac{1}{\lambda}} & \text{otherwise,} \end{cases} \quad (9)$$

is sometimes used. However, this point wise inverse of the Box-Cox transformation is *not* the inverse of the underlying forecast distribution<sup>1</sup>. If only the mean as point estimate is of interest then the *biased adjusted* back-transformed mean can be approximated by [5],

$$y_t \mapsto \begin{cases} e^{y_t} \left[ 1 + \frac{\sigma^2}{2} \right] & \text{if } \lambda = 0 \\ (\lambda y_t + 1)^{\frac{1}{\lambda}} \left[ 1 + \frac{\sigma^2(1-\lambda)}{2(\lambda y_t + 1)^{\frac{1}{\lambda} + 1}} \right] & \text{otherwise,} \end{cases} \quad (10)$$

where by assumption  $y_t$  is the mean on the transformed scale and  $\sigma$  the variance.

**Time series decomposition** We can assume an additive or multiplicative decomposition of a time series,

$$y_t = S_t + T_t + R_t \quad (11)$$

$$y_t = S_t \times T_t \times R_t, \quad (12)$$

where  $S_t$  is the seasonal component,  $T_t$  is the trend-cycle component and  $R_t$  is the remainder component. Note that  $T_t$  contains both, trend and cycle effects. While in financial data analysis the multiplicative approach is common both approaches can be linked by a preceding log transformation (equivalent to a  $\lambda = 0$  Box-Cox transformation). Then a multiplicative ansatz can be translated into an additive one. As a rule of thumb,

- An additive model is appropriate if magnitude of seasonal effects does not vary with level<sup>2</sup>
- A multiplicative model is appropriate if seasonal effects are proportional to the level

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<sup>1</sup>More precisely, the back transformation (10) is only correct for the median if the transformed forecast distribution is symmetric [6]

<sup>2</sup>The level of a time series is the baseline value around which the time series varies.

One common approach to such a decomposition is STL (Seasonal-Trend Decomposition using LOESS).

The *classical decomposition* is the basis of most decomposition methods. The **trend-cycle** component is described by moving averages. An  $m$ -order moving average ( $m$ -MA) is defined for  $m$  *odd* as

$$\hat{T}_t = \frac{1}{m} \sum_{j=-k}^{j=k} Y_{t+j}, \quad j = \frac{m-1}{2}, \quad (13)$$

so essentially it takes the of symmetric ('centered') datapoints in time around  $t$ . Averaging removes some randomness in the data leaving a smooth trend-cycle curve. For  $m$  *even* symmetry is not canonical as one can choose the excess point in time before or after  $t$ . The idea is to take both choices and then center the result by taking the mean,

$$\hat{T}_t = \frac{1}{2m} \left( \sum_{j=-m/2}^{j=m/2-1} Y_{t+j} + \sum_{j=-m/2+1}^{j=m/2} Y_{t+j} \right) \quad (14)$$

$$= \frac{1}{m} \left( \frac{1}{2} Y_{t-m/2} + \frac{1}{2} Y_{t+m/2} + \sum_{j=-m/2+1}^{j=m/2-1} Y_{t+j} \right). \quad (15)$$

So this is essentially the form of an odd moving average where the endpoints have half weight. For the even case the notation  $2 \times m$  - MA is often used to stress symetrization.

The classical decomposition has the following limitations

- The estimate of the trend is not available for the first / last few points in time. This implies limitations on forecasting.
- Seasonal components repeat from year to year, which may not be realistic
- The method is not robust to outliers as it depends on averages.

Since the 1950s there were developed more sophisticated approaches all sharing the disadvantage of being only applicable to monthly and quarterly data.

The *X-11* method benefits from 1) relatively robust to outliers 2) automated choice for seasonal and trend changes and 3) is widely tested on economic data. It has the disadvantage of 1) missing prediction intervals 2) ad hoc model with no underlying model.



Extensions like *X-12-ARIMA* or *X-13-ARIMA* allow for 1) adjustments for trading days and other explanatory variables, 2) known outliers can be omitted 3) level shifts and ramp effects can be modeled 4) missing values can be estimated and replaced and 5.) holiday factors (easter) can be estimated.

The *X-13-ARIMA-SEATS* is model based and was developed for economic data. It allows for 1) smooth trend estimates 2) estimates at end points 3) changing seasonality.

The *Seasonal and Trend decomposition using Loess (STL)* [2] iteratively computes trend-cycle and seasonal components. Loess is a method for estimating nonlinear relationships, which is used for the trend term. STL has the advantage that 1) can handle any type of seasonality 2) seasonal components are allowed to change over time 3) it can be made robust to outliers, however not always improving the results. The disadvantages are 1) no trading day adjustments and 2) only applicable for additive decomposition (which can be incorporated by a log or, more generally, by Box-Cox transformation with  $0 \leq \lambda \leq 1$ )

## Time series features

- Linear trend  $x_t = t$ ,  $t = 1, 2, \dots, T$  where  $t$  is the number of observations until the start of the series. The strong assumption is that the trend continues in the future.
- Dummy variables. Use  $n - 1$  dummy dimensions for  $n$  categories. (if  $n$  dummy dimensions are used then a regression gets singular). E.g. encoding 7 weekdays needs 6 dummy variables. Dummy variables can be also used for outliers, public holidays at varying dates (easter), season, ...etc.
- Intervention, when some external event happened. Can be modelled using dummy variables (spikes). Zero before intervention and 1 otherwise (steps)
- Lagged values. E.g.  $x_1$ : advertising of previous month,  $x_2$ : advertising for two months previously, ...  $x_n$ .
- Fourier series are used to describe seasonal patterns. The Fourier terms

$$s_k(t) = \sin\left(\frac{2\pi kt}{m}\right), \quad c_k(t) = \cos\left(\frac{2\pi kt}{m}\right) \quad (16)$$

can be used as predictors. A regression using these features is called harmonic regression and is of the form

$$y_t = a + bt + \sum_{k=1}^K (\alpha_k s_k + \beta_k c_k) + \epsilon_t,$$

where in this specific case  $a$  is the intercept and a  $bt$  is the regression part on the trend. The value of  $K \leq m/2$  . can be determined by minimizing AICs. Note that  $m$  can be non-integer (weekly data). It is useful for large  $m$ . (For seasonal data  $m = 4$  there is not much benefit and dummies could be used).

## 4 Classical time series models

### Overview of methods

- Regression models
- Exponential smoothing methods
- Arima
- Dynamic Regression
- Hierarchical forecasting
- Vector autoregression
- survival analysis

**Baseline Methods [6].** Let  $\{y_1, \dots, y_T\}$  be historic data. The aim is to predict a future value  $y_{T+h|T}$  based on the historic data. For these models also estimates for the standard deviations  $\hat{\sigma}_h$  are available, which can be used to estimate prediction intervals according to eq. (27). The following baseline models have a prediction distribution that is of the form

$$y_{T+h|T} \sim \mathcal{N}(\hat{y}_{T+h|T}, \hat{\sigma}_h^2) \quad (17)$$

- **Average method:** forecasts future values as mean of historical data,

$$\hat{y}_{T+h|T} = \bar{y} = \frac{1}{T} \sum_{t=1}^T y_t \quad (18)$$

$$\hat{\sigma}_h = \hat{\sigma} \sqrt{1 + \frac{1}{T}}. \quad (19)$$

- **Naïve method / random walk:** forecasts future value as last observed value,

$$\hat{y}_{T+h|T} = y_T \quad (20)$$

$$\hat{\sigma}_h = \hat{\sigma}\sqrt{h}. \quad (21)$$

For financial data in an "efficient market" this is the best forecast (efficient market hypothesis). In practice the naive method works well for stock prices and exchange rates.

- **Seasonal naïve method:** forecasts last observed value of the same season,

$$\hat{y}_{T+h|T} = y_{T+h-np}, \quad (22)$$

$$\hat{\sigma}_h = \hat{\sigma}\sqrt{n}, \quad (23)$$

where  $p$  is the seasonal period and  $n = \left\lfloor \frac{h-1}{p} \right\rfloor + 1$ .

- **Drift method:** adds to naïve method an overall gradient,

$$\hat{y}_{T+h|T} = y_T + \frac{h}{T-1} \sum_{t=2}^T (y_t - y_{t-1}) \quad (24)$$

$$= y_T + \frac{h}{T-1} (y_T - y_1), \quad (25)$$

$$\hat{\sigma}_h = \hat{\sigma} \sqrt{h \left( 1 + \frac{h}{T} \right)} \quad (26)$$

In the estimates for the  $h$ -step standard deviations the quantity  $\hat{\sigma}$  is the empirical standard deviation of the residuals. The average and naïve methods give constant predictions.

The drift method and naïve method increase uncertainty the larger the prediction horizon ( $h$ ). For the prediction intervals, all methods give the same variance for one step ahead predictions  $h = 1$  and large data  $T \rightarrow \infty$ .

**Prediction intervals** Prediction intervals are given by the forecast distribution, in particular its standard deviation. However these are for most models not available and one uses the residual distribution instead. In fact, if the model is parameter-free, then the standard deviations of the model forecast distribution and the residual distribution are identical. If the model has

parameters they would increase the standard deviation (due to uncertainty on the model parameters), which is often ignored.

Further assuming that the residuals are **normally distributed** with constant variance the 95 % one step prediction interval is approximated by  $\hat{y}_{T+1|T} \pm 1.95 \hat{\sigma}$ . Here 1.95 is the 95 %-width of the standard normal distribution.

This can be generalized to multi step prediction interval  $h > 1$ ,

$$\hat{y}_{T+h|T} \pm c \hat{\sigma}_h, \quad (27)$$

where  $\hat{\sigma}_h$  is the (model dependent) standard deviation at step  $h$  and  $c$  is the width of the standard normal distribution for a given prediction interval (e.g.,  $c = 1.95$  for the 95 % prediction interval).

If no analytical standard deviations are available then prediction intervals may be obtained from **bootstrapped residuals**.

Furthermore note that in general prediction intervals are usually too narrow due to unaccounted uncertainties such as model uncertainties or parameter uncertainties.

**Forecasts with transformations.** In general quantiles can be directly back transformed. However, this holds not for the mean. While the canonical way is given by the transformation laws of pdfs there are adjustments for the Box-Cox transformation based on a Taylor series expansion [6], Chap. 5.6.

**Autoregressive models.** Denote the backshift operator  $B$  by  $B(y_t) = y_{t-1}$ . Then differencing can be written as  $(1 - B)$ . An **autoregressive model** of order  $p$  (AR(p)) is of the form

$$y_t = c + \sum_{i=1}^p \varphi_i y_{t-i} + \varepsilon_t = c + \sum_{i=1}^p \varphi_i B^i y_t + \varepsilon_t \quad (28)$$

So this is a multiple regression with lagged values as predictors. For  $p = 1$  there are several limiting cases:

- $c = 0, \varphi_1 = 0$ : white noise ( $\varepsilon$ )
- $c = 0, \varphi_1 = 1$ : naïve method
- $c \neq 0, \varphi_1 = 1$ : drift method
- $\varphi_1 < 0$ :  $y_t$  oscillates around mean.

**Theorem 1** (Stationary autoregressive model [1]). *An autoregressive model remains stationary  $\Leftrightarrow$  the roots of its characteristic polynomial stay outside the unit circle.*

For example, for  $p = 1, 2$  this implies the following constraints on the parameters for a stationary autoregressive model:

- $p = 1$  :  $-1 < \varphi_1 < 1$
- $p = 2$  :  $-1 < \varphi_2 < 1, \varphi_1 + \varphi_2 < 1, \varphi_2 - \varphi_1 < 1$

**Moving average models.**

## 5 Neural networks

**Neural networks.** There are several neural network (NN) based approaches, RNN, LSTM, 1D CNN, deep NN. The basic idea behind deep NN is to translate the problem into a supervised learning problem by predicting  $y_t$  with features that are the last  $l$  target values  $y_{t-1}, y_{t-2}, \dots, y_{t-l}$

**Autoregressive integrated mean average (ARIMA)**

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